# Year 9 Mathematics

## Unit 1: Number

1.2 Calculating and Estimating

# Calculating and Estimating

STARTER · 5 MIN

**Date:** _____ **Title: Calculating and Estimating**

**LEARNING OBJECTIVES**

- Calculate with powers
- Round to a number of significant figures
- Estimate answers by rounding to 1 significant figure

**STARTER** *Answer in your book. No calculators.*

1. Work out $4^2 + 3^2$
2. Work out $-4 \times -4$
3. Round 7.1361 to 1 d.p.
4. Round 506.789 to 2 d.p.

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. **25**
2. **16**
3. **7.1**
4. **506.79**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can calculate with powers | ☐ |
| SILVER | I can round to a number of significant figures | ☐ |
| GOLD | I can estimate answers by rounding to 1 significant figure | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Calculating and Estimating

INTRODUCTION · 3 MIN

Work with powers step by step; round to estimate and check answers.

## Same base

Add or subtract the indices.

$4^4 \times 4^3 \div 4^5 = 4^2$

## Powers of 2

Write 8, 16, 32, 64 as powers of 2.

$8 = 2^3,\ \ 16 = 2^4,\ \ 64 = 2^6$

## Significant figures

Count from the first non-zero digit.

0.025 476 = 0.0255 (3 s.f.)

## Estimate

Round every number to 1 s.f. first.

$38.7 \times 5.2 \approx 40 \times 5 = 200$

# Examples 1 and 2 — Calculating with Powers

EXAMPLES · 5 MIN

BRONZE

*Watch and copy into your book*

**Hint**

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$8 = 2^3$ $16 = 2^4$ $32 = 2^5$ $64 = 2^6$

## Example 1

Work out the value of

$$\frac{7^2 \times 3^4}{3^3}$$

## Example 2

Work out the value of

$$\frac{2^7 \times 8}{16 \times 2^5}$$

# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

**Only the same base combines**

$3^4 \div 3^3 = 3^1 = 3$

$7^2 \times 3 = 49 \times 3$

**= 147**

Tip: leave $7^2$ alone: it has a different base.

## Example 2

**Write 8 and 16 as powers of 2**

$8 = 2^3,\ 16 = 2^4$

$$\frac{2^7 \times 2^3}{2^4 \times 2^5} = \frac{2^{10}}{2^9}$$

**= 2**

Tip: then add and subtract the indices.

# Task 1

PRACTICE · 9 MIN

BRONZE *Work out the value. Show each step.*

**Hint** $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$
$8 = 2^3$ $16 = 2^4$ $32 = 2^5$ $64 = 2^6$

**1** $\dfrac{5^2 \times 2^5}{2^3}$

**2** $\dfrac{3^4 \times 2^6}{3^2 \times 2^4}$

**3** $\dfrac{4^7}{4^5} + 3^2$

**4** $\dfrac{2^6 \times 8}{2^7}$

**5** $\dfrac{27 \times 3^4}{3^5}$

★ $\dfrac{2^5 \times 4 \times 3^2}{8 \times 2^4}$

# Task 1 — Answers

CORRECTIONS

BRONZE

*Tick or correct in green pen*

1 **100**

2 **36**

3 **25**

4 **4**

5 **9**

★ **9**

# Examples 3 and 4 — Significant Figures

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

**Hint** 1st s.f. = first non-zero digit
next digit 5 or more: round up

## Example 3

Round 13 078 to
3 significant figures.

## Example 4

Round 0.254 76 to
2 significant figures.

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

**1st s.f. is 1, 3rd s.f. is 0**

13 078: next digit is 7

so round up

**= 13 100**

Tip: keep the zeros as place holders.

## Example 4

**1st s.f. is 2 (not the 0)**

0.254 76: next digit is 4

so round down

**= 0.25**

Tip: zeros at the start do not count.

# Task 2

PRACTICE · 9 MIN

SILVER

*Round to the number of significant figures given*

**Hint** 1st s.f. = first non-zero digit
next digit 5 or more: round up

1. 48 672 to 2 s.f.
2. 7345 to 1 s.f.
3. 13 078 to 2 s.f.
4. 0.004 862 to 2 s.f.
5. 0.030 59 to 3 s.f.

★ 0.000 704 9 to 3 s.f.

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

1. **49 000**
2. **7000**
3. **13 000**
4. **0.0049**
5. **0.0306**

★ **0.000 705**

# Examples 5 and 6 — Estimating

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

**Hint** round every number to 1 s.f.
volume = length × width × height

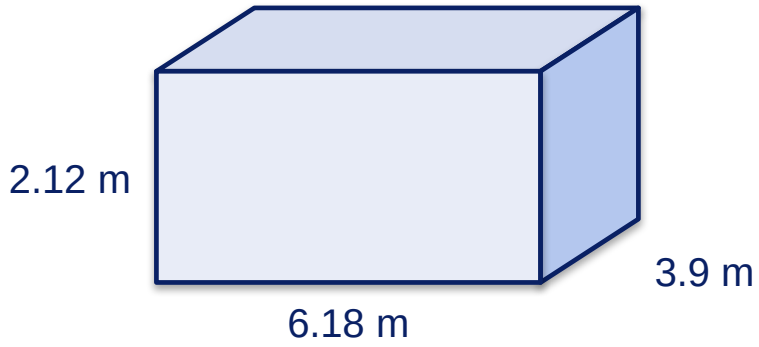
**Example 5** (2 marks)

Estimate

$$\frac{38^2}{4.9} - 9.7^2$$

**Example 6** (2 marks)

Estimate the volume.



# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

**Round each number to 1 s.f.**

$$\frac{40^2}{5} - 10^2$$

$= 1600 \div 5 - 100$

**$= 320 - 100 = 220$**

Tip: round first, then square.

## Example 6

**Round to 1 s.f.: 6, 2, 4**

$6 \times 2 \times 4$

**$\approx 48\ m^3$**

Tip: volume = length × width × height

# Task 3

PRACTICE · 9 MIN

GOLD

*Estimate by rounding each number to 1 s.f.*

**Hint** round every number to 1 s.f.
volume = length × width × height

1. $31.2 \times 4.9$

2. $608 \div 19.7$

3. $\dfrac{6.2 \times 49}{9.8}$

4. $21^2 - 3.9^2$

5. $\dfrac{59^2}{3.1}$

★ Volume of a cuboid
4.1 m by 5.8 m by 2.95 m

# Task 3 — Answers

CORRECTIONS

GOLD

*Tick or correct in green pen*

1. **150**
2. **30**
3. **30**
4. **384**
5. **1200**

★ **72 m³**

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** Work out $\dfrac{3^5 \times 2^3}{3^3}$

**SILVER** Round 0.030 59 to 2 s.f.

**GOLD** Estimate $\dfrac{41^2}{7.9}$

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

**BRONZE** **72**

**SILVER** **0.031**

**GOLD** **200**

# Exam Practice — Calculating with Powers

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: show all your working*

**Hint**

$a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

$8 = 2^3$ $\quad 16 = 2^4$ $\quad 32 = 2^5$ $\quad 64 = 2^6$

## Question 1

(2 marks)

Work out the value of

$$\frac{5^2 \times 2^6}{2^4}$$

.....

## Question 2

(3 marks)

Work out the value of

$$\frac{2^6 \times 8 \times 5^2}{16 \times 2^4}$$

.....

# Exam Practice — Significant Figures

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint** 1st s.f. = first non-zero digit
next digit 5 or more: round up

## Question 3

(2 marks)

Round to 3 significant figures.

**(a)** 40 961 (1)

.....

**(b)** 0.007 049 8 (1)

.....

## Question 4

(1 mark)

A town has 48 672 people.
Round this to 2 significant figures.

.....

# Exam Practice — Estimating

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** round every number to 1 s.f.
volume = length × width × height

## Question 5

(2 marks)

Estimate

$$\frac{31^2}{4.8} + 6.9^2$$

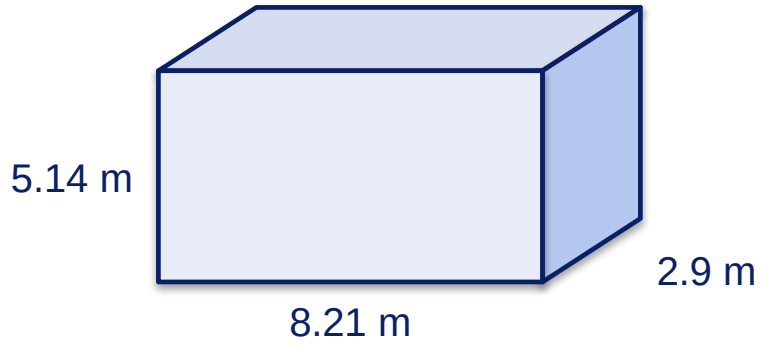
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## Question 6

(2 marks)

Estimate the volume of this cuboid.

*Diagram NOT accurately drawn*




..... $m^3$

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

$2^6 \div 2^4 = 2^2$ — M1

**$25 \times 4 = 100$** — A1

Tip: only the same base combines.

## Question 2

$8 = 2^3,\ 16 = 2^4$ — M1

$2^9 \div 2^8 = 2$ — M1

**$2 \times 25 = 50$** — A1

Tip: write 8 and 16 as powers of 2.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

**(a) 41 000** — B1

**(b) 0.007 05** — B1

Tip: the first s.f. is the first non-zero digit.

## Question 4

**49 000** — B1

Tip: writing just 49 loses the place value.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

$30^2 \div 5 + 7^2$ — M1

**$= 180 + 49 = 229$** — A1

Tip: round to 1 s.f. before squaring.

## Question 6

$8 \times 5 \times 3$ — M1

**$= 120\ m^3$** — A1

Tip: 8.21 → 8, 5.14 → 5, 2.9 → 3